

Thermoregulation and Sweating

Sweating glands and axillary odor (ABCC11, EDAR), hyperhydrosis, heat and cold sensors (TRPM8, TRPV1), heat production (ACTN3, thyroid, brown fat) and malignant hyperthermia

Sérgio F · 03-10-2026 · local time

Informative and educational genetic report, based only on Sérgio F genes (genotyping chip + local imputation). It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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1. Summary

Favourable. Malignant hyperthermia variants are absent (RYR1 R163C and G2434R, CACNA1S R1086H). The heat production is well equipped: DIO2 Thr92/Thr92 (active desiodase), ADRB3 W64/W64 and PGC-1 α G482/G482, without the minor variants. The predisposition to hypothyroidism (percentile 34.3) and hyperthyroidism (38.5) is typical.

To be considered. Thyroid genetic thermostat is deviated: Low genetic TSH (percentile 19.1 score, 8.4 loci) and high free genetic T4 (80.1; 88.5 loci). It is not a disease, but it changes the reading of a thyroid analysis. The excessive sweating loci load is at percentile 82.0, but only with calibrated 3 loci. UCP1 -3826 is heterozygous. In **ABCC11** is C/T: wet cerumen and typical apocrine secretion. **TRPM8** is T/T, the genotype associated with cold adaptation (and also a migraine risk allele), and **ACTN3** is R/X (intermediate cold tolerance).

What does it mean? There are no genetic signs of heat or cold intolerance. The practical points come from the drugs: dapagliflozin, telmisartan and retatrutide increase the risk of dehydration in the heat, in the gymnast and in the exercise.

2. How to read this report

- **Data:** Sérgio F chip genotypes (GRCh38, strand +) and, when SNP is not in the chip, locally imputed genotypes (statistical imputation, public panel of genomes complete to 30x) with quality $DR2 \geq 0.8$. Under this threshold SNP is treated as SNP not found in the data.
- **Key variants** 15 Selected SNPs from the literature (GWAS Catalog, ClinVar, dbSNP and reference studies): 12 of the chip, 2 imputed and 0 not found.
- **Strand:** All genotypes are in the GRCh38 strand +. The published effect allele is compared with the observed alleles; 1 SNPs needed conversion to the complementary strand and 1 are palindromic (A/T or C/G), solved by the reference allele and the position. When the literature uses the strand of the gene, the letters may differ from those of this report.
- **Polygenic scores (PGS):** 4 PGS Catalog scores calculated locally with 895,146 variants in total (chip + imputed). The percentile compares Sérgio F with the reference population ($n = 633$) calculated with exactly the same variants. Percentage ≥ 80 : MORE LIKELY/HIGH; ≤ 20 : LESS LIKELY/LOW; between 20 and 80: TYPICAL.
- **Loci load GWAS:** For each trait, all SNPs with $p \leq 5 \times 10^{-8}$ in GWAS Catalog (1 line for SNP, the most significant) are added as simple count of effect alleles (each allele valley 1). The sum of Sérgio F is compared with that of the reference population ($n = 633$) with exactly the same variants, which gives an empirical percentile. Many of these SNPs are in linkage disequilibrium, so the load is a complement, not a substitute, of the polygenic scores. 2038 loci, 1854 with genotype.
- **Judgment:** Each result is divided into Favourable or Unfavourable (or typical/without judgment). A genotype is a probability, not a destination; almost all effects are small and add to the lifestyle.
- **Evidence:** Strong : replicated and with clear clinical effect; Moderate = GWAS/consistent meta-analysis, modest effect; Weakness : small or mechanistic studies.

3. Polygenic scores

In the scores of hypothyroidism and hyperthyroidism, a higher percentile = more risk. In TSH and free T4 scores, the percentile only says if the level tends to be naturally higher or lower. There are no published polygenic scores for sweating or tolerance to heat or cold.

Thyroid (metabolic thermostat)

Trait	Score (PGS Catalog)	Variants used	Percentile	Result	Judgement
TSH	PGS004906 (Mulder 2023, 60 variants)	57 60 (weighted coverage 95%)	19.1	LOW	without judgment
Free T4	PGS004907 (Mulder 2023, 31 variants)	29 31 (weighted coverage 90%)	80.1	HIGH	without judgment
Hypothyroidism	PGS002024 (Privé 2022, Bayesian method, 633 thousand variants)	621,308 632,597 (weighted coverage 97%)	34.3	TYPICAL	typical
Hyperthyroidism/ thyroidosis	PGS002023 (Privé 2022, Bayesian method, 279 thousand variants)	273,752 279,385 (weighted coverage 94%)	38.5	TYPICAL	typical

4. Key variants by topic

Variants with described effect. Many characteristics of thermoregulation are not good or bad; there the judgment is "without judgment".

Swallow glands, sweat and odor

The eccrine glands produce the sweat that cools the body. The apocrines, in the armpits, produce the secretion that bacteria transform into odor. ABCC11 decides the type of cerumen and apocrine secretion. EDAR affects the density of eccrine glands. Hyperhidrosis has few known loci.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ABCC11	rs17822931	538G>A	C/T	chip	T (1 copy/s)	without judgment	538G>A (G180R; T in strand +): the A allele of the gene gives dry cerumen and almost without axillary odor (reduced apocrine secretion); the G/G type gives wet cerumen and typical odor	Strong
EDAR	rs3827760	V370A	A/A	chip	G (0 copy/s)	without judgment	V370A: increases the density of eccrine sweat glands (common only in some populations)	Strong
PLB1	rs56089836		G/G	imputation DR2=1.00	G (2 copy/s)	unfavourable	Excessive Sewage Locus (phospholipase B1) [GWAS Catalog: Excessive sweating, G allele, OR 1.40, p=2E-11, PMID 29895819]	Weak
LONP2	rs6500380		G/A	imputation DR2=0.97	G (1 copy/s)	unfavourable (light)	Excessive Sewage Locus (LONP2) [GWAS Catalog: Excessive sweating, G allele, OR 1.61, p=7E-10, PMID 29895819]	Weak

Temperature sensors (TRP channels)

The TRP channels are the thermometers of the skin and nerves. TRPM8 detects cold and menthol, and TRPV1 detects heat and capsaicin. Variants of these genes change the sensitivity to cold and thermal pain.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
TRPM8	rs10166942		T/T	chip	T (2 copy/s)	without judgment	TRPM8 (cold and menthol receptor): the frequency of this variant rises with latitude, a sign of adaptation to cold; it is also a locus of migraine (effect allele = risk of migraine) [GWAS Catalog: Migraine, T allele, beta 1.104 [1.09-1.12] increase, p=9E-51, PMID 35115687]	Moderate
TRPV1	rs8065080	I585V	C/C	chip	C (2 copy/s)	without judgment	I585V (C = Val585): heat and capsaicin receptor; V allele is associated with lower sensitivity to cold and thermal pain in some studies	Weak

Heat production: muscle, thyroid and brown fat

The body heat comes from basal metabolism (regulated by thyroid), muscle (tremor and thermogenesis without tremor, where ACTN3 enters) and chestnut fat (UCP1, ADRB3, DIO2, PGC-1 α). The thermogenesis report deepens the brown fat.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
ACTN3	rs1815739	R577X	T/C	chip	T (1 copy/s)	without judgment	R577X: genotype X/X (without α -actinin-3) improves cold tolerance, because the muscle produces more heat without tremor	Moderate
DIO2	rs225014	Thr92Ala	T/T	chip	C (0 copy/s)	favourable	Thr92Ala (C = Ala92): less active 2 deiodinase, less local T3 in muscle and brown fat	Moderate
UCP1	rs1800592	-3826	T/C	chip	G (1 copy/s)	unfavourable (light)	-3826A>G (G in the strand of the gene): minus UCP1 in the brown fat	Weak
ADRB3	rs4994	W64R	A/A	chip	G (0 copy/s)	favourable	W64R (G = Arg64): lower response of the chestnut fat beta-3 receptor	Weak
PPARGC1A	rs8192678	G482S	C/C	chip	T (0 copy/s)	favourable	G482S (T = Ser482): PGC-1 α less active (mitochondria and thermogenesis)	Weak

Malignant hyperthermia and thermal stress

Malignant hyperthermia is a serious reaction to volatile anesthetics and succinylcholine in carriers of RYR1 or CACNA1S variants. The same variants can give severe intolerance to heat and effort. There are more than 40 causal variants; the chip reads only a few.

Gene	SNP	Variant	Genotype	Source	Effect allele (copies)	Judgement	Allele effect	Evidence
RYR1	rs118192172	R614C	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	T faux copy/s)	without data	R614C: pathogenic variant of malignant hyperthermia (volatile anesthetics and succinylcholine) and severe heat intolerance	Strong
RYR1	rs118192161	R163C	C/C	chip	T (0 copy/s)	favourable	R163C: pathogenic variant of malignant hyperthermia	Strong
RYR1	rs121918592	G2434R	G/G	chip	A (0 copy/s)	favourable	G2434R: pathogenic variant of malignant hyperthermia	Strong
CACNA1S	rs1800559	R1086H	C/C	chip	T (0 copy/s)	favourable	R1086H: variant of malignant hyperthermia in the muscle calcium channel	Strong

5. GWAS loci burden

Sum of all loci of GWAS Catalog ($p \leq 5 \times 10^{-8}$) of each trait, with each allele oriented towards the risk (in TSH and in T4 free, towards the highest level), compared to the reference population.

Trait (GWAS Catalog)	Loci (with genotype)	Sérgio F effect alleles	Calibrated variants	Percentile	Judgement
Excessive sweating and hyperhidrosis	5 (3)	3 of 6	3	82.0	unfavourable
TSH (thyroid stimulating hormone)	750 (688)	667 of 1376	686	8.4	without judgment
Free T4	107 (99)	103 of 198	99	88.5	without judgment
Hypothyroidism	1176 (1064)	1016 of 2128	1061	49.1	typical

6. Favourable

- **Malignant hyperthermia:** RYR1 R163C C/C, RYR1 G2434R G/G and CACNA1S R1086H C/C absent. Strong for these variants (not excluding the others).
- **Brown fat and mitochondria:** DIO2 Thr92Ala T/T (Thr/Thr, active deiodinase), ADRB3 W64R A/A and PPARGC1A G482S C/C without the minor function variants. **Moderate / weak.**
- **Typical hypothyroidism and hyperthyroidism:** percentiles 34.3 and 38.5; loci load of hypothyroidism at 49.1. **Moderate.**
- **EDAR V370A absent** (A/A): typical density of sweat glands. Strong (no value judgment).

7. Unfavourable

- **Excessive sweating:** loci load at percentile 82.0; PLB1 G/G in homozygosity and LONP2 G/A with 1 allele. Weak (only calibrated 3 loci).
- **UCP1 -3826 T/C:** 1 allele of minus UCP1 in the brown fat. **Weak.**
- **Low genetic TSH and high free genetic T4** (scores at the 19.1 and 80.1 percentiles): an TSH near the lower limit can be the normal of Sérgio F and should not, alone, lead to changes. **No judgment.**
- **No judgment:** ABCC11 C/T (wet cerumen, typical axillary odor); TRPM8 T/T; TRPV1 I585V C/C (V/V); ACTN3 R577X T/C (R/X). RYR1 R614C is not in the data.

8. Mechanisms of ageing

Mapping in 84 mechanisms of aging of Sérgio F immortality objectives (only the mechanisms affected by this report).

Mechanism	Genetic finding	Meaning
Thermogenesis decline	DIO2, ADRB3 and PGC-1α without variants of minor function; UCP1 -3826 heterozygote	Favourable
Thyroid alterations	TSH genetics in the 19.1 and T4 free percentile in the 80.1; typical hypothyroidism	Neutral
Metabolic Homeostasis Decline	Derived thyroid thermostat for more free hormone; basal metabolism without genetic marker of deficit	Neutral
Mitochondrial dysfunction	PPARGC1A G482/G482 (PGC-1α active)	Favourable
Molecular Platerone Failure	Response to thermal shock (HSP70): no genetic marker read	Neutral

9. Practical suggestions

All suggestions are suggestion (not applied), consistent with the current genetic plan, and should be discussed with the doctor.

1. **Heat and drugs:** dapagliflozin (water loss and sodium in urine), telmisartan and retatrutide (less thirst and less intake) are added to heat, hammam and long exercise. Drinking before having a thirst and replacing sodium on days of very sweat. On days of vomiting, diarrhea or fever, talk to the doctor about temporarily suspending dapagliflozin and telmisartan (the rules of "days of disease"). Suggestion (not applied).
2. **Sauna:** there is no genetic contraindications. Regular exposure to heat activates heat shock proteins, and lunchroom is associated with less cardiovascular mortality in observational studies. With dapagliflozin, sessions of 15–20 min with hydration and without alcohol (already zero). Suggestion (not applied).
3. **Cold:** DIO2, ADRB3 and PGC-1 α are in favor of thermogenesis. If exposure to cold is used, keep it away from strength training, because it reduces hypertrophy. Suggestion (not applied).
4. **Thyroid analysis:** with low genetic TSH and high free genetic T4, a low-normal TSH value with high-normal free T4 is probably constitutional. Always compare with the baseline value before concluding that there is hyperthyroidism. Suggestion (not applied).
5. **Before a general anesthesia:** The three variants of malignant hyperthermia are absent, but this does not exclude the others. A family history of reaction to anesthesia, of high fever without explanation or of rhabdomyolysis with effort should be told to the anesthesiologist. Suggestion (not applied).
6. **Excessive sweating:** if disturbed, the topical aluminum hydrochloride is 1.a line. Avoid oral anticholinergics by the anticholinergic load in the brain. Suggestion (not applied).

Not recommended: Sausage or intense exercise in dehydrated heat or prolonged fasting with dapagliflozin; oral anticholinergics for sweat; thyroid hormone without indication to "accelerated metabolism".

10. Limitations

- Thermoregulation has few genetic studies. There are no polygenic scores for the sweating or tolerance to heat or cold, and the loci load of the sweating has only calibrated 3 loci.
- The chip reads only 3 of the more than 40 variants of malignant hyperthermia (RYR1 and CACNA1S); RYR1 R614C is not in the data. Absence does not exclude susceptibility.
- Heat tolerance depends mainly on acclimatization, physical form, hydration, body composition and drugs.
- The scores of TSH and free T4 were developed in children and explain only part of the variation.
- Conclusions only from genes; it is not a diagnosis.

11. Methods and sources

- **Genotypes:** Sérgio F chip (GRCh38, strand +). Absent SNPs come from local statistical imputation (public panel of complete 30x genomes), used only with $DR2 \geq 0.8$. No genetic data came out of the analysis computer.
- **Alleles and strand:** The alleles of each location are read from the Imputed VCF. The published effect allele is compared with those alleles and, if necessary, converted to the complementary strand (only in non-palindromic SNPs). In palindromic SNPs, strand + (GWAS Catalog standard and the harmonized files of PGS Catalog) is assumed. In the loci of GWAS Catalog with protective effect, the allele of effect is exchanged by the risk allele.
- **Polygenic scores:** Harmonized GRCh38 files from PGS Catalog: PGS004906 and PGS004907 (Mulder 2023), PGS002024 and PGS002023 (Privé 2022). Score = Δ coefficient $\times$ dose of effect allele. The percentile is empirical in the reference population ($n = 633$), with the same variants.
- **Loci load:** GWAS Catalog (associations with $p \leq 5 \times 10^{-8}$), alleles guided by the published effect signal, and empirical calibration in the same reference population.
- **Interpretation:** GWAS Catalog, ClinVar, dbSNP/Ensembl and reference literature: Yoshiura 2006 (ABCC11), Kamberov 2013 (EDAR), Key 2018 (TRPM8), Wyckelsma 2021 (ACTN3 and cold), GWAS of excessive sweating of 2018 and CPIC of 2019 for RYR1 and CACNA1S. The complete references are available on request.

12. Annex: all variants analysed

Topic	Gene	SNP	Genotype	Source	Effect allele	Copies	Chain	Interpretation
Swallow glands, sweat and odor	ABCC11	rs17822931	C/T	chip	T	1	strand +	C/T: wet cerumen (G allele is dominant), typical odor
Swallow glands, sweat and odor	EDAR	rs3827760	A/A	chip	G	0	strand +	0 copy of the effect allele
Swallow glands, sweat and odor	PLB1	rs56089836	G/G	imputation DR2=1.00	G	2	strand + (palindromic; assumed +)	2 risk allele copies
Swallow glands, sweat and odor	LONP2	rs6500380	G/A	imputation DR2=0.97	G	1	strand +	1 copy of risk allele
Temperature sensors (TRP channels)	TRPM8	rs10166942	T/T	chip	T	2	strand +	2 copy of the effect allele
Temperature sensors (TRP channels)	TRPV1	rs8065080	C/C	chip	C	2	strand +	2 copy of the effect allele
Heat production: muscle, thyroid and brown fat	ACTN3	rs1815739	T/C	chip	T	1	strand +	C/T (R/X): intermediate
Heat production: muscle, thyroid and brown fat	DIO2	rs225014	T/T	chip	C	0	strand +	without the risk allele
Heat production: muscle, thyroid and brown fat	UCP1	rs1800592	T/C	chip	G	1	complement (strand)	1 copy of risk allele
Heat production: muscle, thyroid and brown fat	ADRB3	rs4994	A/A	chip	G	0	strand +	without the risk allele
Heat production: muscle, thyroid and brown fat	PPARGC1A	rs8192678	C/C	chip	T	0	strand +	without the risk allele
Malignant hyperthermia and thermal stress	RYR1	rs118192172	SNP not found in the data	SNP not found in the data (DR2<0.8 imputation)	T			SNP not found in the data
Malignant hyperthermia and thermal stress	RYR1	rs118192161	C/C	chip	T	0	strand + (published alleles; absent site of VCF)	without the risk allele
Malignant hyperthermia and thermal stress	RYR1	rs121918592	G/G	chip	A	0	strand + (published alleles; absent site of VCF)	without the risk allele
Malignant hyperthermia and thermal stress	CACNA1S	rs1800559	C/C	chip	T	0	strand + (published alleles; absent site of VCF)	without the risk allele